How fast does the market price in the news?

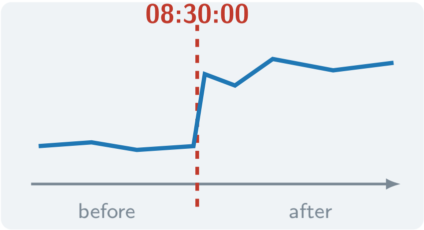
A measurement study. Not a strategy.

`proj-price-discovery` · the model in 14 slides

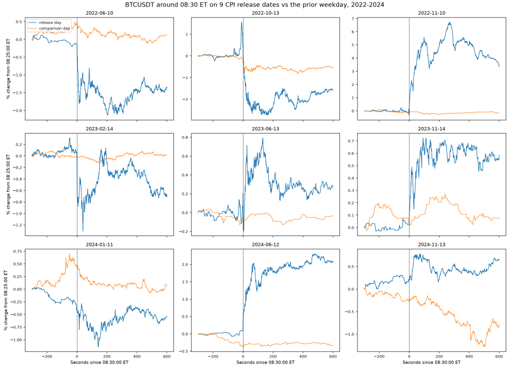
The setup

- CPI lands at **08:30:00 ET**, to the second
- Nobody knows the number one second earlier
- Crypto trades **24/7** — no auction, no open

So the release is a clean shock into a live market.



Does it even react? We checked first.



9 / 9
dates show a jump

5.6×
median vs. a normal day

±
sign follows the surprise

The question, made precise

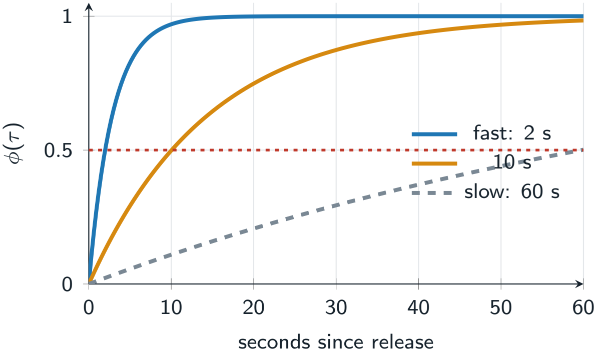
Of the move the release eventually causes,

what fraction has happened by τ seconds?

$$\boxed{\phi(\tau) = \frac{\text{move so far}}{\text{eventual move}}}$$

Report it at 1 s, 10 s, 1 min, 10 min, 1 h. One curve.

What the curve looks like



$$\phi(\tau) = 1 - e^{-\lambda\tau}$$

One number per event: λ.

We report its half-life

$$\tau_{1/2} = \frac{\ln 2}{\lambda}$$

"half the move happened in 6.9 seconds" — a sentence you can argue with.

The obvious way to compute it

For each event, divide. Then average.

big surprise
+40 bp

dull release
+2 bp

noise is ± 3 bp
denominator $\to$ 0

One event can flip the average.

The usual fix is the real trap

"Just drop the events where the price barely moved."

That is selecting on the outcome.

You pick events using information from *after* the release that you never had before it.

It is the fastest route to a clean-looking wrong answer, and the first thing a reviewer will look for.

So we never divide

Give every event **two** parameters instead of one.

A_e amplitude	λ_e rate
how big was the news	how fast it was absorbed
don't care	*the answer*

$$\mathbb{E}\left[R_e(\tau)\right] = A_e\left(1 - e^{-\lambda_e \tau}\right) \quad \Longrightarrow \quad \phi_e(\tau) = \frac{\underline{A_e}\left(1 - e^{-\lambda_e \tau}\right)}{\underline{A_e}}$$

The amplitude cancels on paper. A dull event now gives a *wide* answer, which is the honest response to weak evidence. Nothing is dropped.

214 events. Now what?

No pooling

fit each event alone

214 shrugs

every interval useless

Complete pooling

one λ for all

one number

CPI vs FOMC?

2017 vs 2026?

Partial pooling

own rate, shared family

both

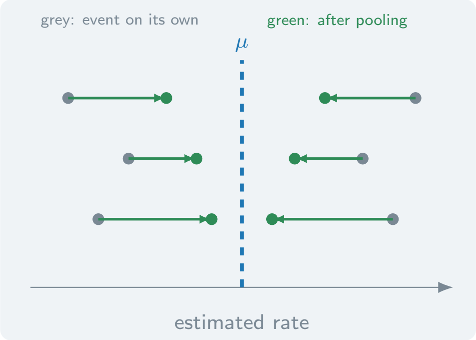
and it self-tunes

$$\log \lambda_e \sim \mathcal{N}(\mu,\ \sigma^2)$$

Every event keeps its own rate — and all the rates come from one family.

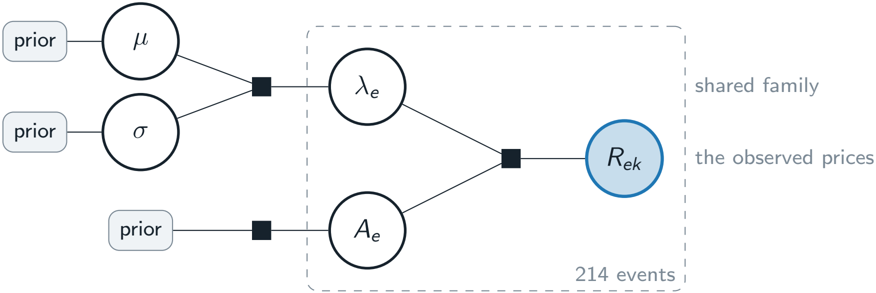
Shrinkage, and why it is not a hack

- Clean event → **keeps its own estimate**
- Noisy event → **drifts to the crowd**
- The weight is not chosen. It falls out of the two variances.



σ **is estimated too.** Events really differ ⇒ σ grows, pooling weakens. Events are alike ⇒ σ shrinks, pooling strengthens. The data decides.

The whole model on one slide



The box repeats per event. Only the shaded node is observed.
The link through μ, σ is how one event's data informs another's estimate.

“Isn’t there just a formula?”

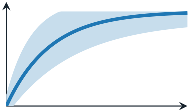
A_e given λ_e plain linear regression	**closed form**
λ_e given A_e λ sits inside $e^{-\lambda\tau}$	**none**
the joint posterior 400+ parameters, no name	**none**

So we sample it.

Hamiltonian Monte Carlo draws from the posterior without ever writing it down.

A 95% interval is literally the 2.5th and 97.5th percentile of the draws.

What we will report



the curve

with a 95% band

$\tau_{1/2}$

one number

half-life in seconds,

with an interval

σ

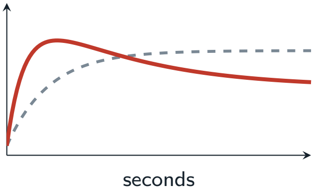
is a finding

large $\sigma \Rightarrow$ there is no

single speed, only a spread

Plus: is CPI faster than FOMC? Has it changed since 2017?
Same model, one extra term.

Where it could break



dashed: what we assume   red: overshoot

- **Overshoot.** One rate cannot describe a spike that reverts. **Two panels of slide 2 already look like this.**
- **The hour contains other news.** We cannot separate it.
- **One venue, one coin.** Binance's clock is Binance's.
- **FOMC is out.** Powell speaks at 14:30, inside the window.

Full list, in a hostile reviewer's words: `docs/limitations.md`

The order matters

The public commit timestamp proving the plan predates the results is the most valuable thing in this repository.